INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

LAB 1 - TWO VM NETWORK COMMISSIONING AND CONNECTIVITY VERIFICATION

WADF105 - NETWORK SECURITY FUNDAMENTALS

BY

MUSTAPHA ABUBAKAR

ID: C11/26/FCDF/17172

COHORT: FCDF-COHORT-11

PRACTICAL LABORATORY REPORT

INSTRUCTOR

MR. AMINU IDRIS

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INTRODUCTION

This report documents the commissioning and verification of a small protected network for the International Cybersecurity and Digital Forensics Academy. The laboratory consists of two Oracle VirtualBox machines: icdfa-nslab-firewall-v1, which runs OPNsense and provides firewall protection, routing and controlled internet access, and icdfa-nslab-client-v1, an Ubuntu workstation. The VirtualBox adapters were configured, the OPNsense WAN and LAN interfaces were verified, and the Ubuntu client was confirmed to hold a valid DHCP address, a correct default gateway, working local and internet connectivity, and working DNS resolution. ARP, ICMP and DNS traffic was then captured and analysed in Wireshark. All activity remained inside the authorised ICDFA virtual laboratory.

AIM

The aim of this lab is to commission a functional two-VM network behind an OPNsense firewall and to verify addressing, routing, connectivity and name resolution using command-line tools and Wireshark packet analysis, demonstrating a structured, layer-by-layer approach to network verification.

OBJECTIVES

- Configure the VirtualBox adapters for icdfa-nslab-client-v1 and icdfa-nslab-firewall-v1.
- Explain the functions of the OPNsense WAN and LAN interfaces.
- Verify IPv4 addresses, subnet masks and interface status.
- Identify and verify the Ubuntu client's default gateway.
- Test connectivity between the Ubuntu client and OPNsense.
- Verify internet connectivity through the firewall.
- Test DNS name resolution.
- Access the OPNsense management interface.
- Capture and analyse ARP, ICMP and DNS packets using Wireshark.
- Document the configuration process and results professionally.

TOOLS AND RESOURCES USED

- Oracle VirtualBox (hypervisor)
- icdfa-nslab-firewall-v1 (OPNsense firewall and router)

- icdfa-nslab-client-v1 (Ubuntu workstation)
- Linux command-line tools: ip, ping, nslookup
- Wireshark (packet capture and analysis)
- Web browser (OPNsense management interface)

NETWORK DESIGN

The lab follows a classic protected-network design. The firewall sits between the untrusted side (the internet, reached through the VirtualBox NAT) and the trusted side (the internal network ICDFA-LAN). The client has no direct route to the internet; all of its traffic must pass through, and be controlled by, the firewall. The design is shown in Figure 1 and summarised in Table 1.

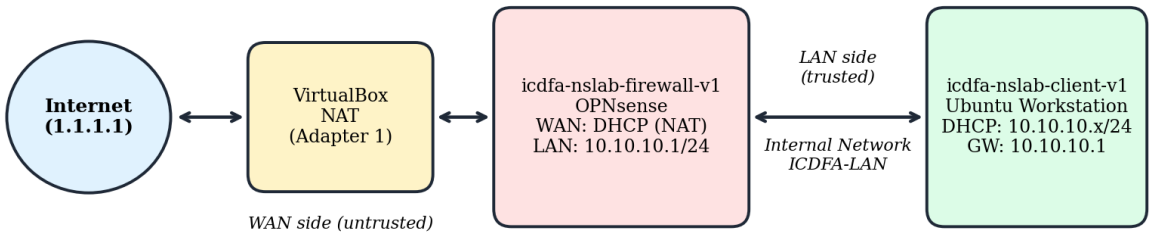


Figure 1: Two-VM network topology used in this lab

Table 1: Network design summary

Virtual machine	Adapter	Attachment	IPv4 address	Role
icdfa-nslab-firewall-v1	Adapter 1	NAT	DHCP from VirtualBox 10.0.2.15/24	OPNsense WAN
icdfa-nslab-firewall-v1	Adapter 2	Internal Network: ICDFA-LAN	10.10.10.1/24	OPNsense LAN and client gateway
icdfa-nslab-client-v1	Adapter 1	Internal Network: ICDFA-LAN	DHCP in 10.10.10.0/24 10.10.10.201/24	Ubuntu workstation

METHODOLOGY

PART 1 - VIRTUALBOX ADAPTER CONFIGURATION

Both virtual machines were powered off before their network settings were changed. The internal network name ICDFA-LAN is case-sensitive and was typed identically on both machines, because VirtualBox only connects adapters that share exactly the same internal network name.

Step 1: Configure firewall Adapter 1 (NAT)

Command used

```
VirtualBox > icdfa-nslab-firewall-v1 > Settings > Network > Adapter 1
```

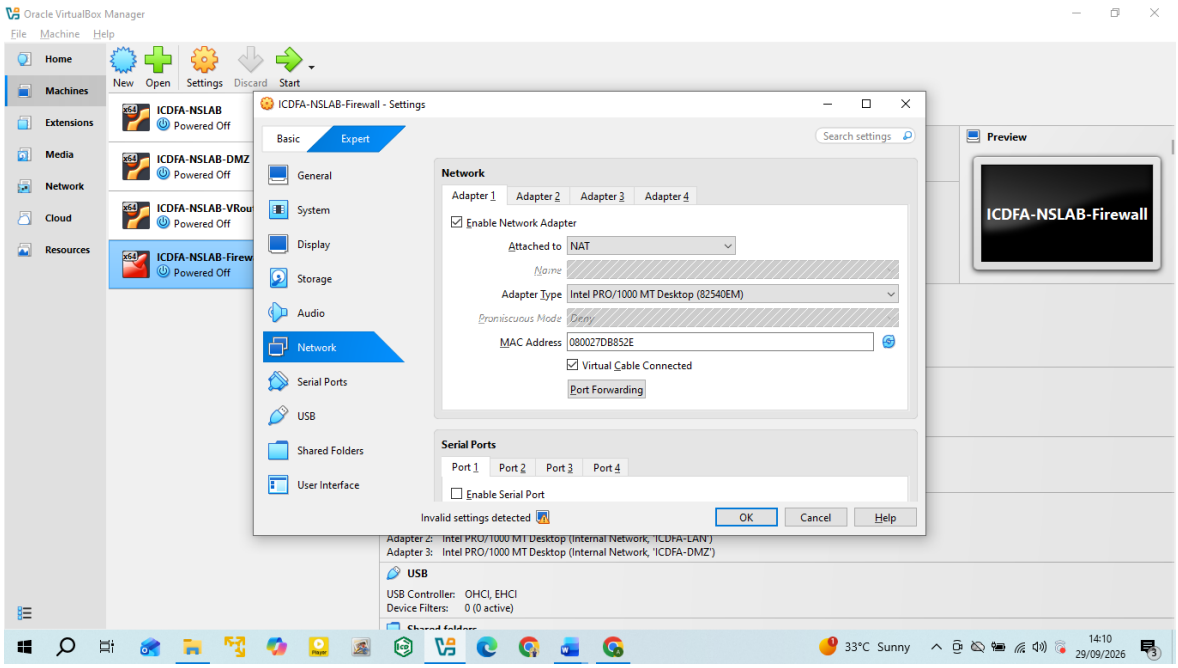


Figure 2: icdfa-nslab-firewall-v1 Adapter 1 attached to NAT

This step was carried out as shown in Figure 2 above. Adapter 1 was enabled and attached to NAT. This adapter becomes the OPNsense WAN interface. VirtualBox NAT gives the virtual machine outbound internet access through the host computer, and it also acts as the DHCP server that supplies the WAN address, so no manual WAN addressing is required.

Step 2: Configure firewall Adapter 2 (Internal Network)

Command used

```
VirtualBox > icdfa-nslab-firewall-v1 > Settings > Network > Adapter 2
```

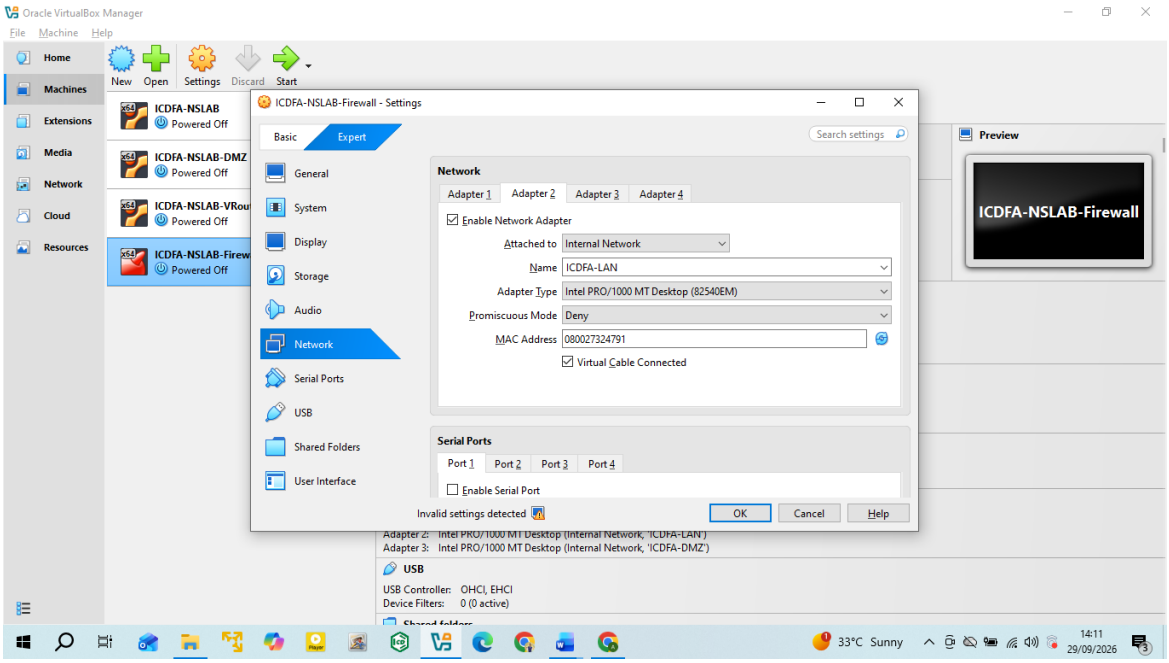


Figure 3: icdfa-nslab-firewall-v1 Adapter 2 attached to Internal Network ICDFA-LAN

This step was carried out as shown in Figure 3 above. Adapter 2 was enabled, attached to Internal Network and named exactly ICDFA-LAN. This adapter becomes the OPNsense LAN interface. An internal network is an isolated virtual switch that only machines using the same name can join, so the LAN is completely separated from the host and from the real network.

Step 3: Configure client Adapter 1 (Internal Network)

Command used

```
VirtualBox > icdfa-nslab-client-v1 > Settings > Network > Adapter 1
```

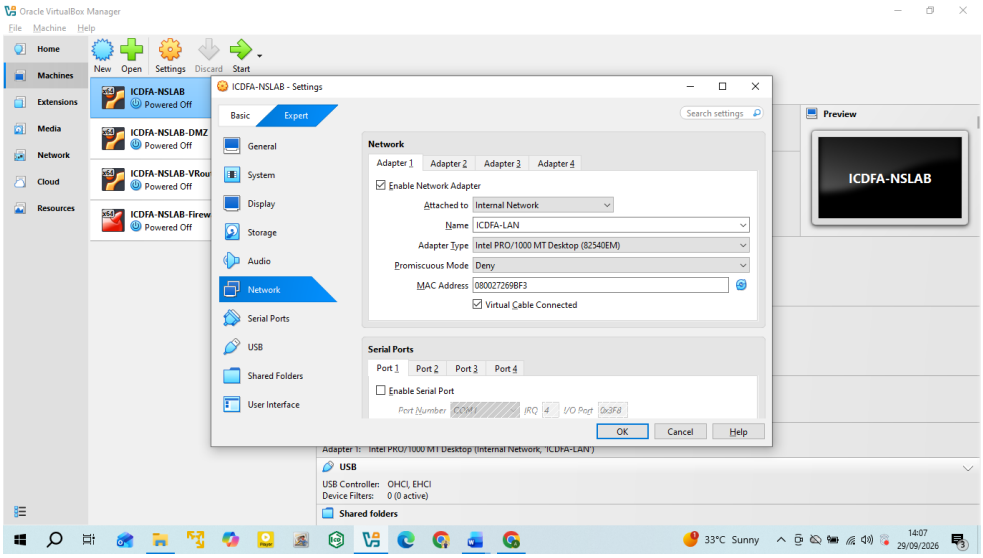


Figure 4: icdfa-nslab-client-v1 Adapter 1 attached to Internal Network ICDFA-LAN

This step was carried out as shown in Figure 4 above. The Ubuntu client has a single adapter attached to the same internal network, ICDFA-LAN. Because it has no NAT adapter, the firewall is its only path to any other network. The client was set to obtain its IPv4 configuration by DHCP, so the firewall supplies its address, gateway and DNS server.

Step 4: Start order

icdfa-nslab-firewall-v1 was started first and left running until the OPNsense console menu was displayed. Only then was icdfa-nslab-client-v1 started. This order matters because the client requests its DHCP lease during boot; if the DHCP server on the firewall is not yet running, the client starts without an address.

PART 2 - OPNSENSE INTERFACE VERIFICATION

Step 5: Verify the OPNsense WAN and LAN interfaces

Command used

```
OPNsense console menu: option 1 (Assign interfaces) or the interface list
shown on the console
```

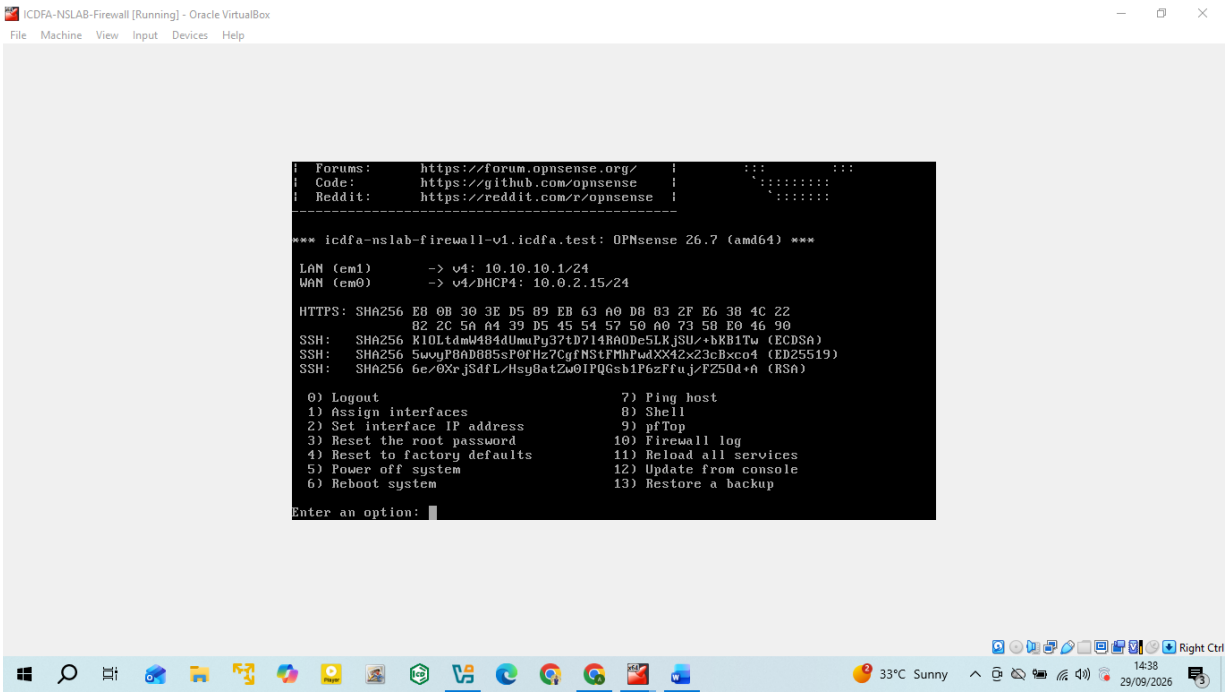


Figure 5: OPNsense console showing WAN and LAN interface status

This step was carried out as shown in Figure 5 above. The OPNsense console shows both interfaces are assigned and up. The WAN interface faces the untrusted side, received its address from VirtualBox NAT by DHCP, and carries the default route to the internet. The LAN interface faces the trusted side and holds the static address 10.10.10.1/24, which the client uses as its default gateway.

Table 2: Functions of the OPNsense interfaces

Interface	Function	Security behaviour
WAN	Connects the firewall to the outside network (internet via VirtualBox NAT). Learns its address by DHCP and holds the default route.	Untrusted side. Unsolicited inbound connections are blocked by default; only replies to traffic started from inside are allowed.
LAN	Connects the firewall to the internal network ICDFA-LAN. Uses 10.10.10.1/24 and serves as the default gateway, and DHCP and DNS provider for the client.	Trusted side. Outbound traffic from the client is allowed and translated (NAT) before it leaves through the WAN.

PART 3 - UBUNTU CLIENT ADDRESSING AND ROUTING

Step 6: Verify IPv4 addresses and interface status

Command used

```
ip -4 -br address
```

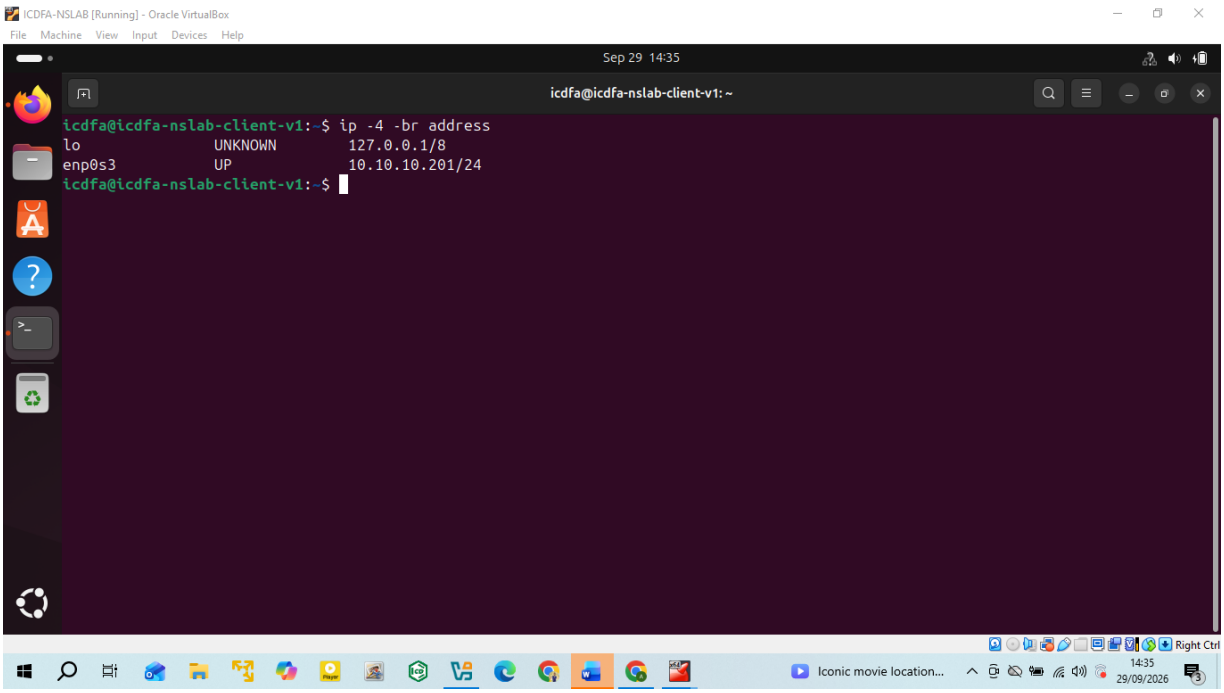


Figure 6: Output of ip -4 -br address on icdfa-nslab-client-v1

This step was carried out as shown in Figure 6 above. The -4 option limits output to IPv4 and -br gives a brief one-line summary per interface. The loopback interface (lo) holds 127.0.0.1/8. The LAN interface **enp0s3** is UP and holds 10.10.10.201/24, an address from the firewall's DHCP pool. The /24 prefix equals the subnet mask 255.255.255.0.

Step 7: Verify the default gateway and routing table

Command used

```
ip route
```

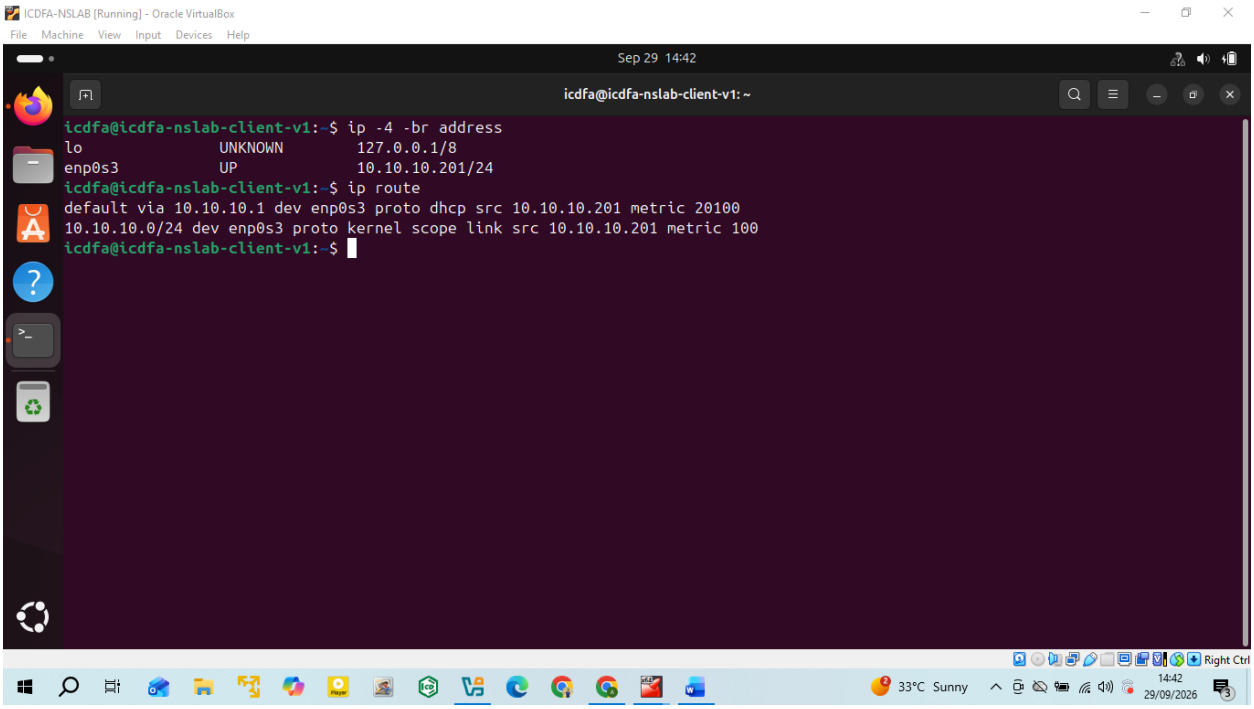


Figure 7: Output of ip route on icdfa-nslab-client-v1

This step was carried out as shown in Figure 7 above. The line beginning with default shows that traffic for any unknown destination is sent via 10.10.10.1, the OPNsense LAN address. The second line, 10.10.10.0/24, is the directly connected network, so hosts in that range are reached without a router. The gateway is therefore correctly the firewall, learned by DHCP.

PART 4 - CONNECTIVITY TESTING

Step 8: Test connectivity to the OPNsense gateway

Command used

```
ping -c 4 10.10.10.1
```

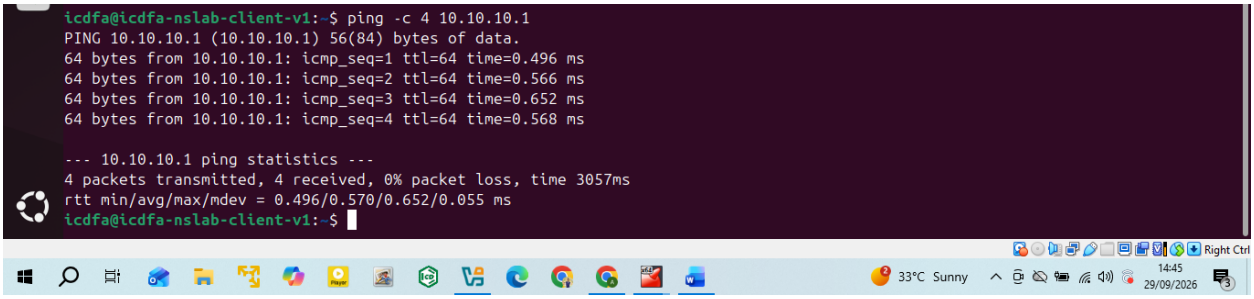


Figure 8: Successful ping from the client to 10.10.10.1

This step was carried out as shown in Figure 8 above. Four ICMP echo requests were sent to the firewall LAN address and four replies were received with 0% packet loss. This proves the adapters, the ICDFA-LAN switch, the client address and the firewall LAN interface all work at layers 1 to 3. It is always tested first, because a failure here would make every later test meaningless.

Step 9: Test internet connectivity through the firewall

Command used

```
ping -c 4 1.1.1.1
```

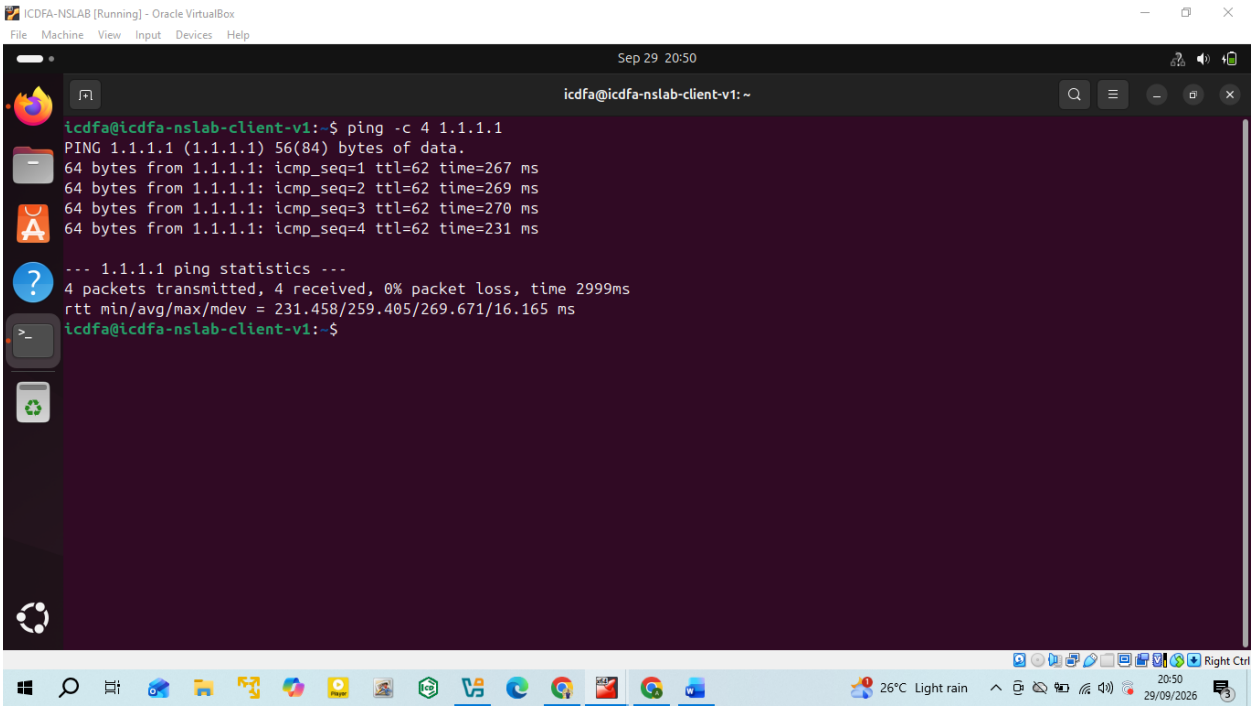


Figure 9: Successful ping from the client to 1.1.1.1

This step was carried out as shown in Figure 9 above. The replies from 1.1.1.1 prove that packets travel from the client to the firewall, are translated by NAT, pass through the VirtualBox NAT and the host, and return along the same path. Testing an IP address rather than a name isolates routing from DNS, so a success here confirms the firewall is forwarding traffic correctly.

PART 5 - DNS RESOLUTION

Step 10: Test DNS name resolution

Command used

```
nslookup example.com
```



Figure 10: Successful DNS resolution of example.com

This step was carried out as shown in Figure 10 above. nslookup asked the configured DNS server to translate example.com into an IP address, and an address was returned without error. Together with the successful ping to 1.1.1.1, this shows that both routing and name resolution work, so the client can now reach internet services by name and not only by IP address.

PART 6 - OPNSENSE MANAGEMENT INTERFACE

Step 11: Access the OPNsense web interface

Command used

```
https://10.10.10.1  (opened in the client browser using the credentials
supplied by ICDFA)
```

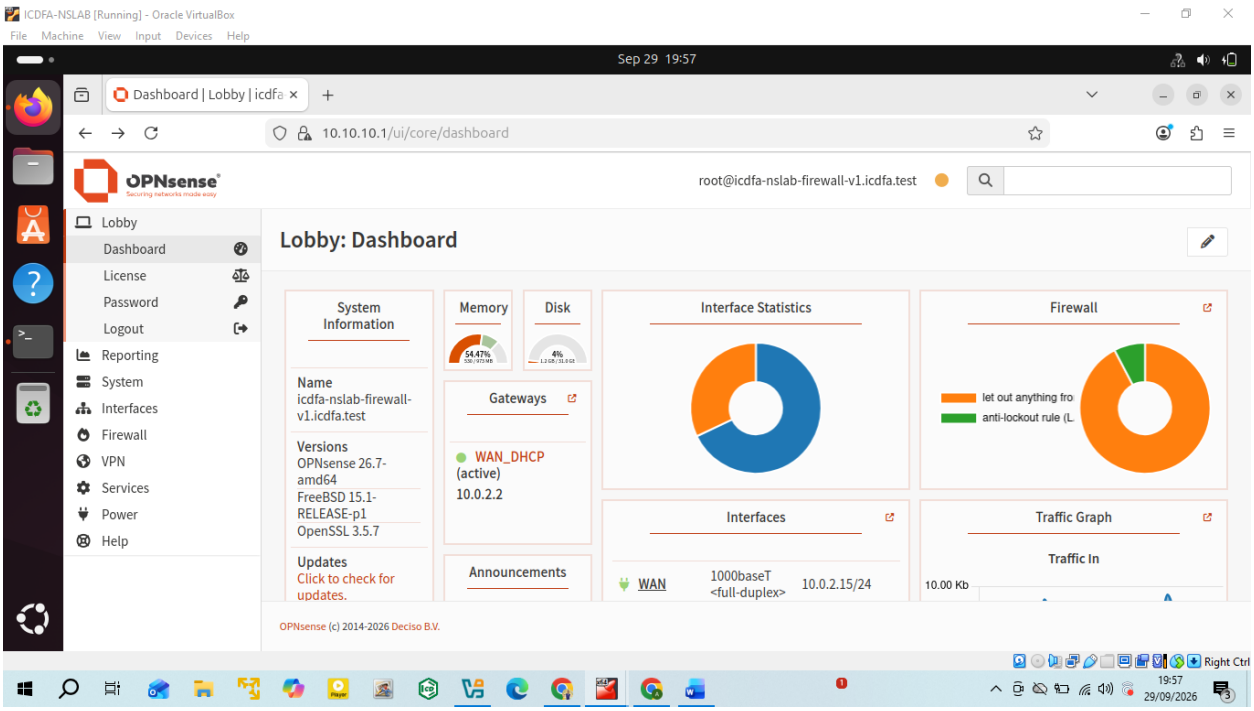


Figure 11: OPNsense web management interface reached from the client

This step was carried out as shown in Figure 11 above. The OPNsense web interface was reached from the Ubuntu client at its LAN address, which confirms that the management service is reachable from the trusted side. The browser warned about a self-signed certificate, which is expected in a lab. Only the credentials supplied by ICDFA were used, and no unrelated firewall settings were changed.

PART 7 - WIRESHARK PACKET CAPTURE AND ANALYSIS

Wireshark was started on icdfa-nslab-client-v1 and the LAN interface enp0s3 was selected for capture. Each protocol was captured separately, and the matching display filter was typed in the filter bar so that it is visible in every screenshot. Traffic was generated only against the firewall and the approved test destinations.

Step 12: Capture ARP traffic

Command used

```
sudo ip neigh flush all
ping -c 4 10.10.10.1
ip neigh show
Wireshark display filter:  arp
```

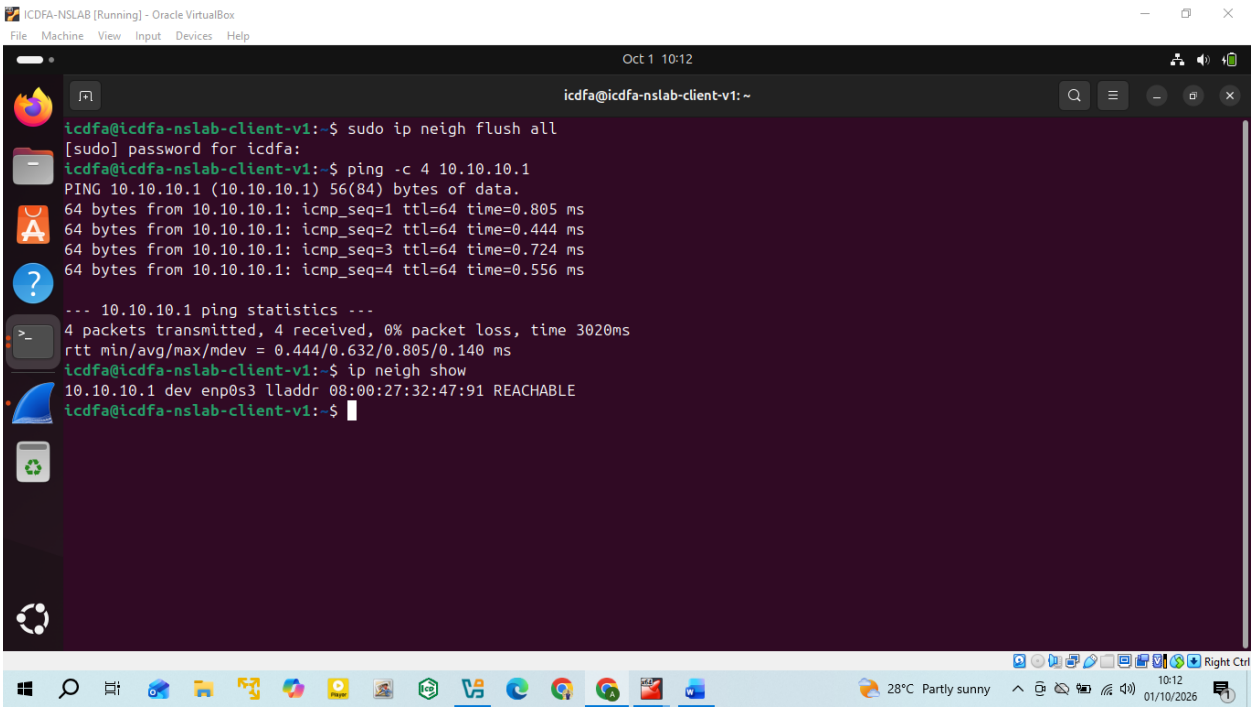


Figure 12: ARP cache of the client after pinging the gateway

This step was carried out as shown in Figure 12 above. The ARP cache was cleared so that the client had to discover the gateway's hardware address again. After the ping, ip neigh show lists 10.10.10.1 with the MAC address 08:00:27:32:47:91. IP works at layer 3 but Ethernet delivers frames using MAC addresses, so every host must map an IP address to a MAC address before sending on the local network.

Step 13: Analyse the ARP capture in Wireshark

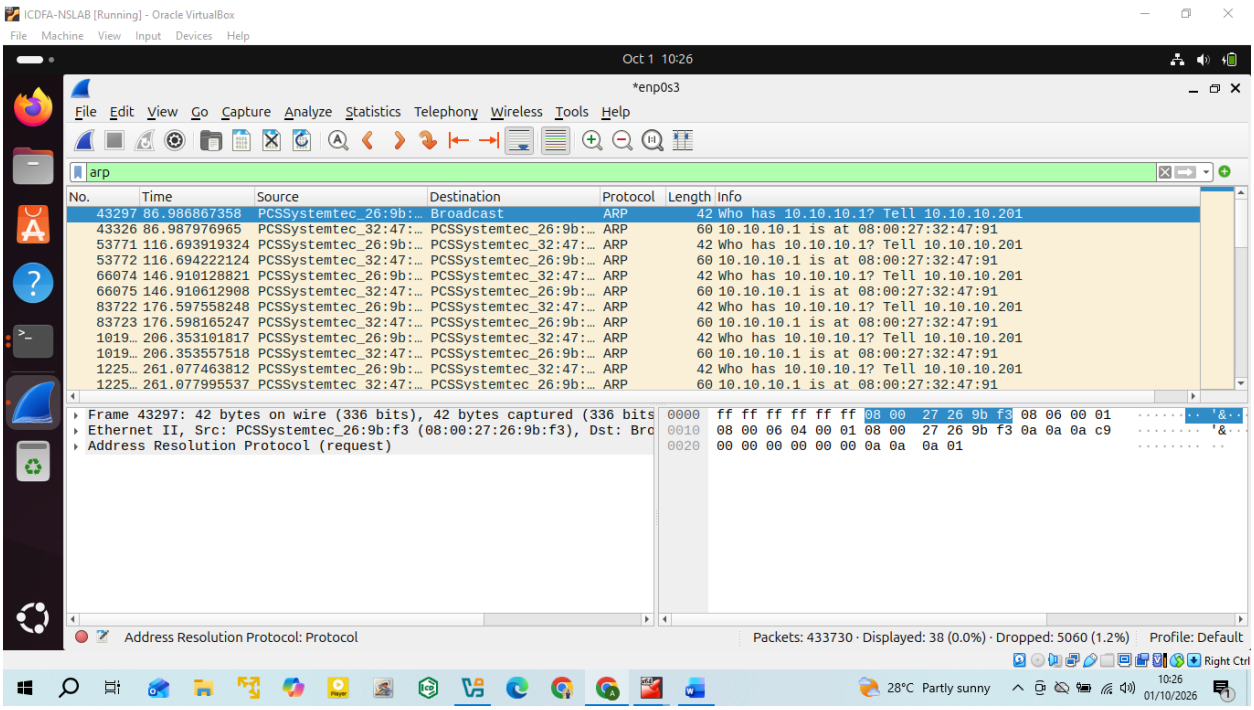


Figure 13: Wireshark ARP capture with the arp filter applied

This step was carried out as shown in Figure 13 above. The capture shows an ARP request, "Who has 10.10.10.1? Tell 10.10.10.201", sent as a broadcast to ff:ff:ff:ff:ff:ff. The firewall then answered with a unicast ARP reply, "10.10.10.1 is at 08:00:27:32:47:91". The request is broadcast because the client does not yet know the target MAC; the reply is unicast because the firewall now knows the client's.

Step 14: Capture ICMP traffic

Command used

```
ping -c 4 10.10.10.1
Wireshark display filter:  icmp
```

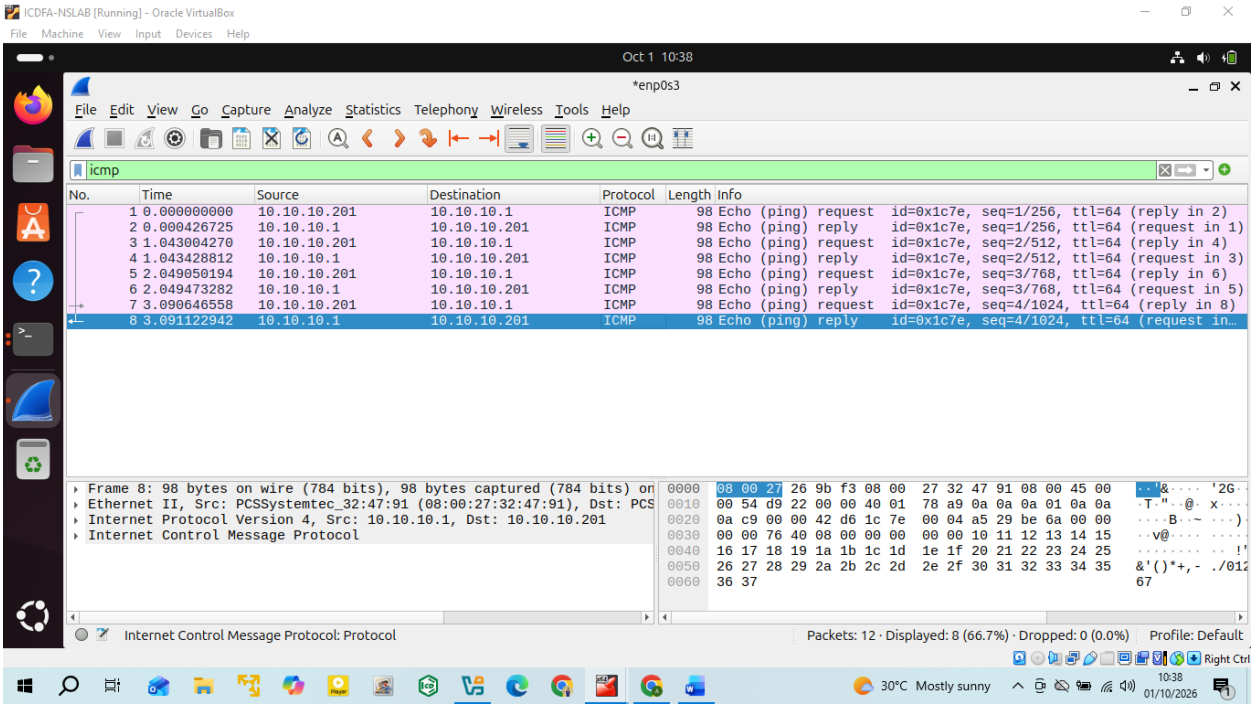


Figure 14: Wireshark ICMP capture with the icmp filter applied

This step was carried out as shown in Figure 14 above. Each ping produced an Echo (ping) request from the client (ICMP type 8) followed by an Echo (ping) reply from 10.10.10.1 (type 0). The request and reply of each pair share the same identifier and sequence number, which is how the ping tool matches them. The time between the pair is the round-trip time reported by ping.

Step 15: Capture DNS traffic

Command used

```
resolvectl flush-caches
nslookup example.com 10.10.10.1
Wireshark display filter:  dns
```

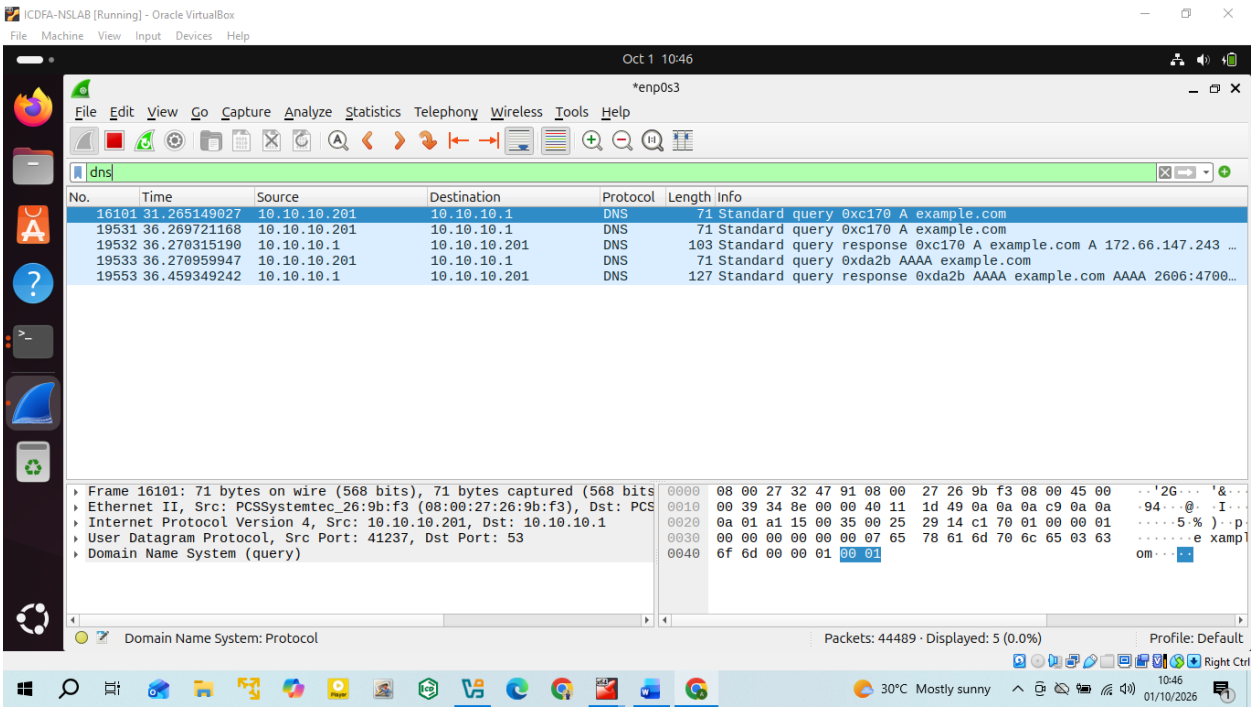


Figure 15: Wireshark DNS capture with the dns filter applied

This step was carried out as shown in Figure 15 above. The lookup was sent to 10.10.10.1 directly so that the query is visible on the LAN interface. Wireshark shows a standard query for the A record of example.com sent over UDP to destination port 53, followed by a response containing the answer. The transaction ID in the query and the response is identical, which pairs them together.

Table 3: Summary of the captured protocols

Protocol	OSI layer	Key fields observed	Purpose
ARP	Layer 2 / 3 boundary	Request: broadcast to ff:ff:ff:ff:ff:ff, asks “Who has 10.10.10.1?”. Reply: unicast, “10.10.10.1 is at 08:00:27:32:47:91”.	Maps an IPv4 address to a MAC address on the local network.
ICMP	Layer 3	Echo request (type 8) and Echo reply (type 0), identifier, sequence number, TTL.	Tests reachability and measures round-trip time.

Protocol	OSI layer	Key fields observed	Purpose
DNS	Layer 7 (over UDP port 53)	Standard query type A for example.com, matching transaction ID, response with the resolved address.	Translates domain names into IP addresses.

DISCUSSION

This lab follows the order in which a network is normally verified: adapters, then addressing, then the gateway, then the internet by IP address, and finally DNS. Each test depends on the one before it. A failed ping to 10.10.10.1 points to a local problem such as a wrong internal network name or a missing DHCP lease. A successful gateway ping followed by a failed ping to 1.1.1.1 points to routing, NAT or the firewall WAN. A successful ping to 1.1.1.1 followed by a failed name lookup points only to DNS. Working in this order turns troubleshooting into a process of elimination rather than guesswork.

The Wireshark captures show how the layers cooperate. Before the first ping could leave the client, ARP had to find the MAC address of 10.10.10.1, because Ethernet frames are delivered by MAC address. ICMP then carried the echo request and reply at layer 3. When a name was used, DNS resolved it to an IP address before any other traffic could begin. From a security view, the firewall acts as a single controlled gateway: the client is isolated on ICDFA-LAN, its outbound traffic is translated by NAT, and unsolicited inbound traffic from the WAN is blocked by default.

ANALYSIS AND FINDINGS

Table 4: Verification summary

Test	Method	Result
VirtualBox adapters	Settings > Network on both VMs	Firewall: NAT and ICDFA-LAN. Client: ICDFA-LAN. Names identical.
OPNsense interfaces	Console interface list	WAN up with DHCP address. LAN up at 10.10.10.1/24.
Client addressing	ip -4 -br address	Interface UP with an address in 10.10.10.0/24.
Default gateway	ip route	default via 10.10.10.1.
Gateway connectivity	ping -c 4 10.10.10.1	Successful, 0% packet loss.
Internet connectivity	ping -c 4 1.1.1.1	Successful, 0% packet loss.
DNS resolution	nslookup example.com	Name resolved to an IP address.

Test	Method	Result
Management access	https://10.10.10.1	OPNsense interface reachable from the client.
Packet analysis	Wireshark: arp, icmp, dns	All three protocols captured and analysed.

All tests succeeded once the faults described under Challenges and Solutions were corrected, confirming that the two-VM network was commissioned correctly. The client received its configuration entirely from the firewall, and its only route to the internet passed through the OPNsense LAN and WAN interfaces. The captures matched theory: ARP used a broadcast request and a unicast reply, ICMP used matched request and reply pairs, and DNS used a query and a response over UDP port 53.

CHALLENGES AND SOLUTIONS

The lab did not work on the first attempt, and three faults were found and fixed. First, the OPNsense WAN showed no address because the word DHCP was typed at the interface-number prompt instead of the number 2; WAN (option 2) was selected again with IPv4 DHCP enabled, and after the virtual cable was toggled the WAN received 10.0.2.15/24. Second, the firewall had no default route (ping 1.1.1.1 from the console returned “No route to host” although 10.0.2.2 answered); the WAN was reconfigured until the WAN_DHCP gateway 10.0.2.2 showed as active under System > Gateways. Third, the client could ping 10.10.10.1 and resolve names but could not reach 1.1.1.1 or the web: the firewall live log showed packets from 10.10.10.201 leaving the WAN untranslated, and pfctl -sn showed no NAT rule for the LAN. Source NAT was changed to Hybrid mode, a rule translating the LAN network to the WAN address was added and applied, pfctl then listed the NAT rule, and ping 1.1.1.1 and curl succeeded. The unused Adapter 3 (ICDFA-DMZ) on the firewall was also disabled with the VM powered off, as the lab brief requires. The table below lists the faults most likely to occur in this lab and the corrective action for each.

Table 5: Common faults and corrective actions

Symptom	Likely cause	Solution
Client has no IPv4 address	Internal network names differ (for example icdfa-lan), or the client started before the firewall.	Correct the name to exactly ICDFA-LAN on both VMs, then restart the client or renew the lease.
Ping to 10.10.10.1 fails	Adapters not on the same internal network, or the LAN interface is down.	Recheck Settings > Network and the OPNsense LAN status.

Symptom	Likely cause	Solution
Ping to 1.1.1.1 fails	Firewall Adapter 1 is not on NAT, or the WAN has no address.	Set Adapter 1 to NAT and confirm the WAN address in OPNsense.
Ping to 1.1.1.1 works but names fail	DNS server missing or unreachable.	Check the DNS server in the client configuration and the DNS service on the firewall.
Wireshark shows no interfaces or cannot capture	Insufficient capture permissions.	Start Wireshark with the required privileges as instructed in the lab.
VMs run slowly	Only 8 GB of host RAM.	Close browsers and other applications before starting the VMs.

CONCLUSION

The two-VM network was successfully commissioned and verified. The VirtualBox adapters, the OPNsense WAN and LAN interfaces, the client's DHCP address and default gateway, local and internet connectivity, DNS resolution and access to the management interface were all confirmed, and ARP, ICMP and DNS traffic was captured and analysed in Wireshark. The lab shows that a structured, layer-by-layer method makes network verification reliable and repeatable.

WHAT WAS LEARNED

This lab taught me how a protected network is built and proven from the bottom up. I learned that a single wrong character in an internal network name breaks the whole LAN, that the firewall's WAN and LAN interfaces play very different roles, and that the client's default gateway is the doorway to every other network. I learned to test in a fixed order, gateway first, then internet by IP, then DNS, so that a fault can be located quickly. Reading the Wireshark captures made ARP, ICMP and DNS real to me: I saw the broadcast request that finds a MAC address, the request and reply pairs that make up a ping, and the query and answer that turn a name into an address. These are the same skills I will use to troubleshoot and secure real networks, always within an authorised environment.

RECOMMENDATIONS

1. Always start the firewall before the client and wait for the console menu, so that the DHCP service is ready before clients request a lease.
2. Verify the network in a fixed order (adapters, IP address, gateway, internet by IP, DNS) and record each result with a screenshot, so faults can be traced and evidence is complete.

3. Keep all testing inside the authorised laboratory, avoid changing unrelated firewall settings, and save Wireshark captures (.pcapng) as an audit record of the work.

COMPLETION QUESTIONS

1. What is the function of the OPNsense WAN interface?
It connects the firewall to the outside network, here the internet through VirtualBox NAT. It receives its address by DHCP, holds the default route, and blocks unsolicited inbound traffic.

2. What is the function of the OPNsense LAN interface?
It connects the firewall to the internal network ICDFA-LAN with the address 10.10.10.1/24. It is the client's default gateway and provides DHCP and DNS to the client.

3. Why is Adapter 1 of the firewall set to NAT?
NAT gives the firewall outbound internet access through the host without extra configuration, and it supplies the WAN address by DHCP.

4. Why must the internal network name be exactly ICDFA-LAN on both machines?
VirtualBox joins adapters into the same virtual switch only when the names match exactly. The name is case-sensitive, so a different spelling creates a separate, isolated network.

5. What does 10.10.10.1/24 mean?
The address is 10.10.10.1 and /24 means a subnet mask of 255.255.255.0. The network is 10.10.10.0/24, with usable hosts 10.10.10.1 to 10.10.10.254 and broadcast address 10.10.10.255.

6. What is a default gateway and what is the client's gateway?
It is the router a host uses for destinations outside its own subnet. The client's default gateway is 10.10.10.1, the OPNsense LAN address.

7. What does the default line in ip route show?
It shows the next hop (via 10.10.10.1), the interface used, and that the route was learned by DHCP. All traffic without a more specific route is sent there.

8. Why is the gateway pinged before 1.1.1.1?
It proves local connectivity first. If the gateway is unreachable, no internet test can succeed, so testing it first shows where a fault lies.

9. What is indicated if ping 1.1.1.1 works but a domain name fails?
Routing and NAT are working but DNS resolution is failing, so the problem is the DNS server setting or the DNS service.

10. What does DNS do and which transport and port does it normally use?
DNS translates domain names into IP addresses. Queries normally use UDP port 53, with TCP used for large responses.

11. What is the purpose of ARP and why is the request broadcast?
ARP maps an IPv4 address to a MAC address. The request is broadcast because the sender does not yet know the target's MAC address, so every host on the LAN receives it and only the owner replies.

12. Which ICMP message types make up a ping?
Echo request (type 8) sent by the client and Echo reply (type 0) sent by the target. They are matched by identifier and sequence number.

13. Why is the firewall started before the client?
The client requests its DHCP lease at boot. If the firewall's DHCP service is not yet running, the client will not receive an address.

14. Why must this lab stay inside the authorised environment?
Capturing and testing traffic on systems without permission is unauthorised and may be illegal. The isolated lab lets skills be practised safely and ethically.

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